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Influences of Organizational Culture and Climate on Individual Creativity

ABSTRACT This paper provides a framework for conceptualizing and reviewing the literature on the influences of organizational culture and climate on individual creativity. Although often treated interchangeably, culture and climate are distinct constructs operating at different levels of meaning; yet at the same time, they are closely interrelated. Culture is the beliefs and values held by management and communicated to employees through norms, stories, socialization processes, and observations of managerial responses to critical events. The beliefs and values that typify a culture for creativity become manifested in organizational structures, practices, and policies. In turn, these structures, practices, and policies guide and shape individual creativity by creating a climate that communicates both the organization's goals regarding creativity and the means to achieve those goals. The paper concludes with a discussion of issues relating to the development of cultures and climates for creativity and potential new directions for future research.

INTRODUCTION The important role of creativity in organizations is attracting an increasing amount of attention from both practitioners and researchers. In business journals and books, practitioners continuously emphasize the need for organizations to create conditions that support creativity as a means for promoting organizational effectiveness, and even survival, in the face of fierce competition and rapidly changing technological, economic, regulatory, and market conditions (e.g., Loeb, 1995; Verespej, 1994). Researchers have responded to calls for examining the innovation process in organizational settings by developing models of creativity and innovation (e.g., Amabile, 1988; Woodman, Sawyer, & Griffin, 1993). Because individuals are the source for the development of new ideas that are the essence of creativity, much of this work has focused on what motivates or enables individual creativity (Redmond, Mumford, & Teach, 1993; Scott & Bruce, 1994). The role of various aspects of the organizational environment and how they relate to individual creativity are receiving increasing attention. In particular, organizational culture (e.g., Kanter, 1988) and climate (e.g., Mumford & Gustafson, 1988) have been identified as important contextual factors that influence individual creativity. However, the terms culture and climate are often used interchangeably in the organizational creativity and innovation literature (Kabanoff & Rossiter, 1994). Although closely related and interdependent, culture and climate are in fact distinct constructs that, when considered in combination, can specify with some precision the contextual environment that impacts behavior in organizations (Reichers & Schneider, 1990; Schneider & Rentsch, 1988). Recent reviews have noted that these concepts are promising areas for understanding creativity in work settings and have called for additional research integrating the two constructs (e.g., Anderson & King, 1993; Kabanoff & Rossiter, 1994). However, there have been few attempts to bring these two important constructs together to explore how they function jointly to shape and guide individual creativity in organizations. This paper is intended to help fill this gap.

Organizational creativity and innovation are closely related sets of activities that involve the development of new ideas with subsequent application (Mumford, this issue). Based on the work of a number of authors (e.g., Amabile, 1988; Kanter, 1988; Woodman et al., 1993), the creativity and innovation process can be considered as involving multiple facets. Creativity begins with problem recognition which leads to the generation of novel ideas, products, services, or processes by an individual or group of individuals. At this stage the process shifts from one of generation of a new idea or solution to implementation within the larger organization. This requires organizational innovation which involves further development of novel ideas and putting them to use in the organization. Although recognizing

that organizational creativity and innovation are closely intertwined, our focus is on relationships between organizational culture and climate and creativity. One reason for this is that a wide range of organizational attributes in addition to culture and climate are necessary for effective organizational innovation (i.e., for testing, implementing, and assessing creative new ideas). These attributes have been specified in detail elsewhere (e.g., Arad & Hanson, this issue; Tornatzky, Fleischer, & Chakrabarti, 1990). Relatedly, translating individual creativity into innovation and effectiveness at the organizational level involves a complex series of linkages that are dependent on a wide range of organizational functions (e.g., financing, marketing) and strategic choices (see Harris, 1994, for a discussion of the organization-to-individual linkage problem). Thus, we concentrate on organizational culture and climate conditions that foster individual creativity, while recognizing that many other components need to exist for individual creativity to become translated into effectiveness through innovation.

In light of this distinction, the goals of this paper are threefold. First, we offer a theoretical framework that describes how organizational culture and climate operate together in conjunction with organizational structures, most notably, human resource practices, to form a larger organizational context for behavior in organizations. Second, using this framework, we review the extent to which culture and climate — as distinct, but interrelated constructs — may influence creativity in work settings. Third, based on this review and integration, some directions are suggested for future research and practice for understanding and developing a culture and climate for creativity in organizations.

AN INTEGRATED FRAMEWORK FOR UNDERSTANDING RELATIONSHIPS BETWEEN ORGANIZATIONAL CULTURE, CLIMATE, AND INDIVIDUAL BEHAVIOR

Before considering how organizational culture and climate relate to the creativity process, it is necessary to first introduce these concepts, discuss their differences and interrelationships, and describe how they influence individual organizational behavior. This discussion serves to set the stage for applying culture and climate to the study of creativity and for integrating the literature on culture, climate, and creativity into a unified framework.

Researchers have studied organizational culture from a number of different perspectives by studying the meaning conveyed in beliefs and values (Schein, 1985), behavioral norms and expectations (Cooke & Rousseau, 1988), stories (Martin, Feldman, Hatch, & Sitkin, 1983), rites and ceremonies (Trice & Beyer, 1984), and organizational structures (Kerr & Slocum, 1987). Each of these approaches describes elements of an organization's culture that operate at different levels of meaning (Rousseau, 1990; Schein, 1990). The deepest layers of culture contain the basic beliefs and values that represent the things that are taken for granted as individuals conduct their business in the organization and define what types of behaviors are considered appropriate. In other words, these aspects of culture represent the shared assumptions held by individuals as they cope with the external and internal tasks that face the organization. These often originate from company "founders" whose own beliefs, values, and assumptions provided a basic model of how the organization should operate and be structured. Beliefs and values become enacted and articulated by subsequent leaders (a) through what management pays attention to, measures, and controls; (b) based on how leaders respond to critical situations and crises; (c) as a result of the types of criteria emphasized in the reward, promotion, recruitment, and selection policies; and (d) through leaders' managerial styles and patterns of behavior (Schein, 1990).

Thus, an "outward press" is exerted by cultural beliefs and values and is reflected in the basic patterns of activity (e.g., decision making, planning) which form the organization's structures and enable it to solve basic problems, such as dealing with the external environment and coordinating internal business operations and human resource functions (Cooke & Rousseau, 1988). As the beliefs, assumptions, and values that represent the deeper levels of culture are put into practice by management, they become articulated in the form of organizational policies, practices, procedures, and systems (Schein, 1990). Thus, how an organization structures and supports the efforts involved in creating a new product, service, idea, process, or procedure will reflect these cultural values.

In our conceptualization, organizational structures, practices, policies, and procedures that are enacted by management and reflect the fundamental values of the organization's culture play a critical role in communicating to employees *what* is being emphasized in the organization. By structures, practices, policies, and procedures we refer to any set of organizational activities or arrangements that may in some way influence how individuals go about doing their work. As examples, they may include patterns of managerial behavior, financial and accounting practices, human resource policies and procedures, or physical work arrangements. Because human resource practices such as recruitment, selection, reward, training, and evaluation are organizational practices specifically intended to manage

employee motivation and work behavior, we focus most of our attention on these types of activities as they relate to organizational culture and climate characteristics that encourage creativity.

The effects of human resource practices on individual work-related attitudes and behaviors are well documented (e.g., Guzzo, 1988). Certainly, some of these effects on individuals operate directly. For instance, by selecting those who have a creative orientation, are highly skilled, and have high ability, or by providing training in creative decision making, it is likely that employees will demonstrate greater creativity for the basic reason that essential knowledge and skill sets that lead to creativity have been enhanced (Woodman et al., 1993). However, many of the influences that human resource practices have on individuals' attitudes and behaviors operate in a more indirect fashion by signaling to those in the organization what is important (Kopelman, Brief, and Guzzo, 1990).

These shared perceptions held by members of the organization about its policies, practices, and procedures are typically defined as climate (Schneider, 1990). As described by Kopelman et al. (1990, p. 295), "climate is the psychological process that mediates the relationships between the work environment (conceived as an objective set of organizational policies, practices, and procedures) and work-related attitudes and behaviors." As a perceptual medium, climate indicates to people in the organization the types of goals that are important and the way in which those goals may be accomplished. Different organizational goals and imperatives can be associated with specific types of climates (Schneider, 1990; Schneider & Rentsch, 1988). Climate answers the question, "What is important around here?" — be it customer service (Schneider & Bowen, 1985), safety (Zohar, 1980), or innovation (Abbey & Dickson, 1983; Nystrom, 1990; Scott & Bruce, 1994). By defining what types of objectives are considered important (e.g., developing new services before competitors) and how they are to be accomplished (e.g., introducing new ideas through the organization's idea suggestion program), organizational climate serves as an important influence on particular behaviors of interest, such as creativity.

In light of this discussion, then, organizational culture and climate are considered as related, but distinct constructs. As noted by Reichers and Schneider (1990):

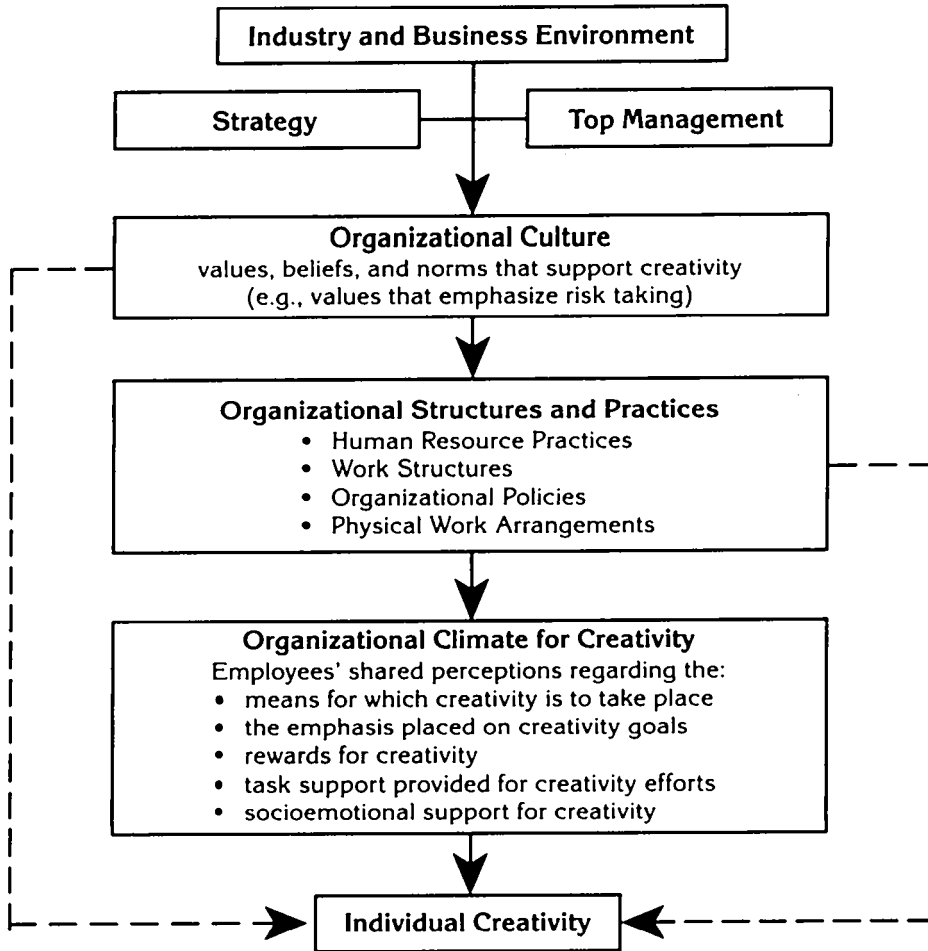
Both climate and culture deal with the ways by which organization members make sense of their environment. These sense-making attempts manifest themselves as shared meanings that form the basis for action. Both climate and culture are learned, largely through the socialization process and through symbolic interaction among group members. (p. 29)

Despite their similarities and the fact that both concepts have been used to describe the context that influences the behavior of individuals in organizations, notable differences exist. The most critical distinction is that culture operates at a greater level of abstraction than climate (Reichers & Schneider, 1990; Schneider & Rentsch, 1988). While climate refers to organization members' shared perceptions of policies and procedures, culture represents the basic values and assumptions that *underlie* those policies and procedures (Reichers & Schneider, 1990; Schneider, Gunnarson, & Niles-Jolly, 1994; Schneider & Rentsch, 1988).

This interrelationship between climate and culture is depicted in the model of the relationship between organizational culture, climate, and individual creativity presented in Figure 1. Figure 1 reflects the above discussion and follows from previous work linking culture, climate, and individual behavior (e.g., Kopelman et al., 1990; Schneider & Rentsch, 1988). The basic elements of organizational culture (the shared assumptions, behavioral expectations, beliefs, and values held by members of the organization) are considered to influence individual creativity in two ways. First, individuals come to understand the values, expected behaviors, and social knowledge through various forms of socialization, such as direct interactions with co-workers and supervisors (Chatman, 1991; Louis, 1980). Culture can be transmitted more indirectly through vicarious observations of the types of behaviors and actions that are sanctioned and rewarded by management (Thompson & Luthans, 1990). Through activities enacted by management and formal and informal socialization processes, norms become established that are learned by and shared amongst employees about appropriate behaviors and how things in the organization should operate. Norms serve as important sense-making devices that can guide and shape behavior (Louis, 1980).

Second, in a more indirect fashion, the basic assumptions, beliefs, and values of the organization's culture become enacted in established forms of activity and behaviors and are reflected in its structures, policies, practices, and procedures. Certain types of structures, policies, and practices can directly impact creativity in the workplace, such as resource support to pursue the development of new ideas. However, these practices, procedures, and management systems, when considered together,

FIGURE 1. A model of the influences of organizational culture and climate on individual creativity.



also influence how individuals interpret their work environment in terms of the types of goals (e.g., developing new products) they see as being emphasized by the organization, and, thus, activate and guide behavior. For instance, organizations with strong cultural values that support creativity are likely to have reward, training, performance management, job design, and other related human resource systems that facilitate risk-taking and experimenting, promote cooperative and interdependent behavior, and support a high tolerance for ambiguity and unpredictability (Schuler & Jackson, 1987).

INFLUENCES OF ORGANIZATIONAL CULTURE ON CREATIVITY

The following section expands the model presented in Figure 1 by reviewing the literature on the influences of organizational culture and climate on individual creativity. First, we briefly describe how conditions in the organization's larger environment, strategic orientation, and top management can function to foster a culture for creativity. Next, we discuss the key assumptions and values that characterize organizational cultures that support creativity. Finally, we conclude this section by reviewing how these key assumptions and values, operationalized in the form of organizational practices, policies, and procedures, produce a climate that communicates the organization's goals for developing new and innovative products or solutions and how these goals are supported and can be achieved.

Industry
Environmental,
Strategic, and Top
Management
Influences on
Culture for Creativity

Certain environmental conditions, strategic approaches, and top management values and actions are associated with cultures that support creativity. Kanter, (1988) noted that certain characteristics of an organization's larger economic and competitive environment serve as "fertile fields" for innovation. Industry environments characterized as highly complex and dynamic (e.g., continuous change in products, technology, and consumer preferences) provide more opportunities for a strategic orientation and culture supportive of innovation to be successful (Gordon, 1991). For instance, in a study analyzing the strategic responses of different hospitals to a city-wide doctor's strike, Meyer (1982) found that the hospital where top management viewed the strike as an opportunity to experiment with new administrative practices reacted more innovatively and handled the strike more successfully than did the hospitals where management took a more conservative and reactive approach. The ways in which top managers interpret and react to crises and events such as this serve as powerful forces that develop norms about appropriate behavior and communicate to employees the values and assumptions about how the organization should function (Schein, 1990). Responses to critical incidents form the basic and enduring elements of the organization's culture that become a shared vision for those in the organization that can help guide responses to ill-defined, complex, and uncertain events that will arise in the future.

Meyer's (1982) analysis provides a demonstration of how top level management in successful innovative organizations, either through official corporate policy or more informally, can instill strong values, beliefs, and assumptions that encourage creativity. The beliefs and values of managers themselves are critical and are often associated with the level of organizational innovation (e.g., Hage & Dewar, 1973; Pierce & Delbeque, 1977). Values and beliefs may be communicated by top management in the form of the organization's mission statement (Amabile, 1988). More important to the development of a culture that supports creativity and innovation, however, is not what management says, but what it *does* (Schneider et al., 1994). The values, beliefs, and basic assumptions of the founder and subsequent leaders are instilled by what leaders pay attention to and emphasize and how they react to critical situations (Schein, 1990). Top managers influence the creativity process by embedding key assumptions, beliefs, and values in established patterns of behavior and interaction that define the organization's culture (Redmond et al., 1993).

A telling example of how top leaders' values and beliefs become established in a culture that nurtures creativity is Ciotta's (1987) description of the development of the corporate culture of Rich Products, a frozen foods producer. A fundamental belief of the company's founder — that new and successful products are only developed by taking risks and making a series of initial failures — has become firmly established in Rich's culture. This value is communicated to Rich's employees through a set of stories, each describing how a new and highly successful product was developed based on a high-risk decision that initially failed. The most famous of these stories involves the genesis of the company when its founder, by accidentally freezing a soybean-based whipping cream replacement, discovered the first non-dairy whipped topping and, thus, launched the frozen non-dairy products industry. Other similar critical incidents in the history of the organization are told in the form of stories that describe the development of new products. By the sharing and repeated telling of these stories, the values and beliefs (i.e., risk taking, continuing in the face of failure) they contain become strongly associated with the management of the organization and the organization itself.

A critical way in which organizational factors can support the creativity and innovation process is by focusing attention on opportunities and the need for new ideas, products, processes, or solutions (Kanter, 1988; Mumford & Gustafson, 1988). At the organization level, Amabile (1988) describes this as the basic orientation of the organization to produce the motivation to innovate and argues that it is critical in generating the intrinsic motivation necessary for the creative process to begin. Because the creativity process begins by recognizing the need for novel problem solutions, features of organizational culture that facilitate such recognition are essential. Top management often plays an important role at this stage. As Amabile (1988) notes, a critical role of top management early in the creativity process is in finding opportunities to reveal basic values and assumptions — such as perceiving events and situations in the organization's environment as opportunities for innovation and communicating this message effectively to the rest of the organization. Management can nurture cultures that encourage the search for new opportunities by instilling values and a normative system that support a diversity of beliefs, free exchange of information, open questioning, and change (Amabile, 1988; Kanter, 1988; King & Anderson, 1990; Woodman et al., 1993). With the use of stories (such as the one described earlier about the founder of Rich Products), metaphors, symbols, and concepts, organiza-

tional leaders can focus attention on searching for and recognizing new opportunities by continuously challenging employees to reexamine what they typically take for granted (Nonaka, 1991).

In addition to focusing attention and motivation on new opportunities for creativity, features of organizational cultures that promote creativity and innovation provide individuals with a strong sense of personal efficacy. Kanter (1988) discusses this as a "culture of pride" typical of organizations that see themselves as leaders in their field and place high value on the capability of its members. A culture that communicates to employees that they are highly competent can serve to build general self-esteem and may even enhance feelings of task-specific self-efficacy, an individual attitude associated with creativity (Farr & Ford, 1990; Mumford & Gustafson, 1988; Redmond et al., 1993; Woodman et al., 1993).

Several of these relationships were identified in one of the few empirical studies on culture and climate and creativity. In this study, Nystrom (1990) examined innovation across four divisions of a Swedish chemical manufacturing firm. Although Nystrom used the term *innovation*, it was broadly defined in terms of new product development, and thus might be better described as creativity (i.e., idea generation) than innovation (i.e., idea implementation). What he found were large differences among the four divisions in the extent to which their cultures and climates supported creativity. These differences were related to the strategic orientation, structure, leadership, and rate of new product development of the divisions. Members of the division at the less innovative end of the continuum described the basic values, normative beliefs, and assumptions of their division as focused on profit and survival, with little value placed in risk taking, challenge, and the intrinsic enjoyment of work. In contrast, the culture of the most creative division emphasized values of change, risk-taking, and competitiveness, while other values, such as customer orientation, quality, efficiency, and profitability were considered less important.

Despite the qualitative nature of the study and the lack of a large sample size, some important points emerge from Nystrom's (1990) analysis. First, a high level of creativity depends on the *configuration* of entrepreneurial leadership, a corporate strategy geared toward seeking new opportunities and markets, a structure that is flexible and responsive to customer needs, and a culture and climate that encourage creativity. Additional research is needed to understand more specifically how these organizational factors operate in tandem to produce an environment that is supportive of creativity. Second, creativity-enhancing cultures may support modes of thinking and behaving that are detrimental to other valued outcomes, such as short-term profitability and quality. This may appear to conflict with statements from other researchers that successful organizations with innovative cultures have strong orientations toward service excellence and quality (e.g., Schneider et al., 1994; West, 1990). However, extremely strong values for creativity that include risk taking, rapid decision making, competitiveness, and experimentation make it difficult to focus the attention on details of the production process that require continuous, but slow, incremental improvements. The notion of *incremental innovation*, involving careful measurement, observation, and testing in an effort to reduce variation and improve quality, is more consistent with strategies geared toward quality and service than a notion of innovation as radical or dramatic change (Anderson, Rungtusanatham, & Schroeder, 1994). In fact in Nystrom's (1990) study, the second most creative division displayed a culture supportive of incremental innovation, by encouraging creativity, but also having a strong customer orientation and encouraging internal cooperation. Third, it is important to consider the types of personal characteristics that enable individuals to thrive in a creativity-enhancing culture. Another potentially negative characteristic of the most innovative division in Nystrom's study was that members experienced a lack of role clarity and had to demonstrate a good deal of self-initiative. Research in the area of person-organization fit (e.g., Chatman, 1991) suggests that individuals who lack the values, beliefs, and traits that have been found to be related to creativity (e.g., persistence, energy, need for autonomy, broad interests; Woodman et al., 1993) will be less successful in these environments. Thus, these individual traits are not only beneficial in terms of their direct relevance to creative behavior, but also in terms of facilitating the degree that new organizational entrants fit into and adjust to a creativity-enhancing culture.

The fundamental values, beliefs, and assumptions that are the basic elements of culture are reflected in observable patterns of behavior, organizational systems and procedures, organizational structure and design of work, and the physical design of the work environment. Schein (1990) describes these more tangible aspects of the organization as the means of *articulation and reinforcement* of the underlying values, beliefs, and basic assumptions that characterize the organization. Hence, in organizations with strong cultural values for creativity, it is likely that one would observe (a) patterns of behavior that demonstrate high levels of activity, emotional involvement in work, excitement about the

importance of work, and at least moderate levels of confrontation and debate in meetings; (b) organizational systems and practices that promote risk-taking, challenge, inventiveness, and growth; (c) an organizational and work structure that encourages communication and personal autonomy; and (d) a configuration of the physical environment that connotes equality and allows for frequent interaction.

The next section focuses on how these organizational practices, structures and work designs, and arrangements of the physical environment, serve to communicate and reinforce norms and values by creating a climate for creativity. A culture for creativity exists when employees adopt basic beliefs (e.g., risk taking is important) and assumptions (e.g., when encountering a new situation, use it as an opportunity to experiment) that are similar to those held by management. However, these values and assumptions, even if strongly held, are rather amorphous. It falls on organizational practices and activities to more directly operationalize the beliefs and assumptions of the culture in a way that communicates to employees the organization's goals and how those goals are achieved, rewarded, and supported by the organization.

ORGANIZATIONAL STRUCTURES AND PRACTICES AND CLIMATE FOR CREATIVITY

Climate provides a basis for interpretation, identifies important goals and the means to achieve them, and creates a force for action (Kopelman et al., 1990; Schneider & Rentsch, 1988). A climate for creativity, then, concerns the extent to which individuals perceive that organizational policies, practices, and procedures specify goals related to the development of new products, ideas, services, or processes and the means to function creatively. The limited amount of research in-

vestigating a climate for creativity and innovation has provided few clear associations linking practices and structures with perceptions of climate. Because we believe that this is in part due to the lack of a theoretical framework that accurately depicts the climate construct, we describe a climate for creativity in more specific terms than has been done previously. We also draw associations between specific climate dimensions and organizational policies, practices, and procedures that relate to creativity.

Given that climate, whether it be for customer service, quality, or creativity, is based on shared perceptions of organizational members of their work environment, and that these environments contain various policies, practices, and procedures, it follows that climate is a multidimensional construct. Building on prior conceptualizations of climate (e.g., Campbell, Dunnette, Lawler, & Weick, 1970; Payne & Pugh, 1976), Kopelman et al. (1990) have specified a core set of five dimensions that apply to a variety of work settings. These dimensions have been adapted for use in describing the construct of organizational climate for creativity and are listed in the right column of Table 1. In the left column are examples of organizational structures, practices, policies, and procedures that relate to each of these five dimensions of climate for creativity. Some of these aspects of the work environment are likely to be associated with more than one of these dimensions of climate (Kopelman et al., 1990). For instance, formal training in creative decision making sends a message that the organization is encouraging creativity in peoples' work behaviors (i.e., how work should get done or means emphasis), as well as providing the necessary skills to perform in a more creative manner (i.e., the tools to get the work done, or task support). However, the advantage to considering climate for creativity in multidimensional terms is that it permits describing how particular aspects of the work environment (e.g., leadership and supervision, training, compensation practices), through perceptions of climate dimensions, relate to important aspects of motivation that encourage and support creative behaviors. Some examples of practices thought to relate most directly to particular climate dimensions are considered in greater detail below.

In order for climate to be supportive of creativity, it is important that outcomes such as the development of new products or services are communicated to employees as goals that are valued by the organization. When employees perceive that management values goals related to creativity, a *goal emphasis* element of a creative organizational climate is said to exist. A climate with a strong goal emphasis is important because it means employees are aware of the standards that can be used to evaluate performance and that they have a basis with which to judge how they are doing. A strong goal emphasis can also help facilitate the develop of self-efficacy expectations regarding their own creative behavior, which are important for generating and sustaining motivation, effort, and persistence toward task accomplishment (Bandura, 1986). Perceptions of efficacy regarding innovation and change have been found to be important in the development of new ideas, products, and services (Farr & Ford, 1990; Mumford & Gustafson, 1988; Redmond et al., 1993; Woodman et al., 1993).

As several researchers have described (Ekvall & Tangeberg-Andersson, 1986; Kanter, 1988; West, 1990), innovative organizations have clearly specified objectives or missions that are widely shared by

TABLE 1. Examples of Organizational Practices, Policies, and Procedures and Related Dimensions of Climate for Creativity

Examples of Supportive Practices, Policies, and Procedures	Corresponding Climate for Creativity Dimension
— clear articulation of goals for the development of new products, services, and processes by top management — communication of goals to employees (e.g., mission statements)	Goal emphasis — the extent that goals for creativity and innovation and the standards for achieving those goals are made known to employees
— managerial and supervisory practices that encourage viewing problems from multiple perspectives (e.g., use of cross-functional teams) — employee involvement practices that promote extensive participation (e.g., self-managing teams, suggestion programs, problem solving teams)	Means emphasis — the extent that the methods and procedures for creativity and innovation are conveyed to employees
— reward and recognition practices that encourage intrinsic motivation — evaluation of new ideas/suggestions based on the contribution of idea, not the creator's place in the organizational hierarchy	Reward orientation — the extent that rewards and evaluations are allocated on the basis of creativity and innovative results
— adequate supply of materials, time, money for innovation efforts — training to develop critical knowledge and skills that support creative behaviors — structures providing employees technical and financial support for new ideas (e.g., idea development centers)	Task support — the extent that employees believe that they are being supported by allocations of the time, funding, equipment, materials, and services necessary to function creatively and to implement new ideas, projects, and solutions
— managerial/supervisory styles that encourage risk taking and active collaboration and provide interpersonal support — evaluation policies that do not punish well-intentioned actions that result in failures, but instead encourage learning from failures	Socioemotional support — the extent that employees believe that the work environment provides the interpersonal support necessary to feel free to function creatively

Note. The basic climate dimensions are adapted from Kopelman et al. (1990).

members of the organization. Stemming from basic principles of goal setting (Locke, Shaw, Saari, & Latham, 1981), these need to be challenging, yet realistic and attainable; specific and well-defined; and attractive or valued (West, 1990). More specific to the unique requirements for encouraging creativity, Amabile (1988; Amabile & Gryskiewicz, 1987) suggests that goal setting should be more tight and specific at the level of missions and outcomes, but should be more loose and flexible in terms of the methods and procedures used to reach these goals. The overall vision for creative behavior developed by management needs to be reinforced and communicated further by actual demonstrations of management's commitment to and sponsorship of creativity and innovation by providing emotional and financial support for new ideas, products, or services (Schneider et al., 1994).

Also important for developing self-efficacy expectations and, hence, motivation and effort, is that employees recognize how to go about performing their work. The *means emphasis* element of a climate for creativity communicates the message of how employees can function creatively and strive for goals relating to change. Whereas an emphasis on strict adherence to rules and procedures and close observation from superiors frustrates attempts at innovation and change, a work environment that allows active risk taking, permits the possibility of errors, shares new insights, and opens debate and dialog to develop creative ideas to their potential signals and encourages the types of behaviors and

actions that lead to creativity (West, 1990). Several authors have noted that a managerial style can emphasize how to be creative by consistently encouraging employees to challenge basic assumptions regarding their work, the products and services they produce, and the processes used to produce them, and encouraging them to address problems from different approaches (Amabile & Gryskiewicz, 1987; Couger, Higgins, & McIntyre, 1993; Nonaka, 1991; Redmond et al., 1993). Because supervisors are particularly salient representatives of management policies, procedures, and actions that in turn can influence climate perceptions (Kozlowski & Doherty, 1989), a supervisory style that encourages subordinate creativity can have a significant impact on the development of such a climate.

Given that greater autonomy and decision making latitude stemming from participative leadership enhances creative behavior (Farr & Ford, 1990; Frischer, 1993; King & Anderson, 1990; Peltz & Andrews, 1966; Redmond et al., 1993), it is not surprising that an active emphasis on employee involvement practices has been associated with creativity. West (1990) stresses the importance of employees recognizing that their input on new ideas and change is encouraged, and actively supported, by management. Subordinates who are provided more decision latitude, autonomy, and influence from their supervisors tend to see their organization as more supportive of innovation (Scott & Bruce, 1993). Because high-involvement approaches encourage problem-solving activity to take place throughout the organization, more opportunities are created for new solutions and ideas to be developed by the people who work directly with customers and products (Lawler, 1992).

Organizational practices, policies, and procedures that recognize and acknowledge the creative contributions of employees provide a *reward orientation*. Following an expectancy theory framework (e.g., Porter & Lawler, 1968; Vroom, 1964), a strong reward orientation for innovation stems from an evaluation and reward structure that signals and reinforces high performance and creative behavior. Outcome expectations for attempting to change existing practices, present a new product, implement a different process, or other instances of creativity, can be influenced in a variety of ways including financial rewards, promotions, formal or informal recognition, increased authority, or enhanced feelings of self-worth and competence (Farr & Ford, 1990). Methods of reward and recognition must be planned carefully, however, so as to maintain employees' intrinsic motivation. Amabile (1988) notes that employees are less creative when motivated primarily by money or recognition, especially when they are striving toward goals set by others. Workers can be rewarded for creative performance as long as care is taken not to contaminate their self-motivation (Couger, Higgins, & McIntyre, 1993). Several authors have reported how organizations with highly creative climates have managed to design recognition and reward practices that encourage employees to develop new ideas (e.g., Erez, 1992; Nonaka, 1991).

Organizational practices, policies, and procedures that emphasize *task support* communicate that the organization is supplying the tools to allow employees to function creatively in their work activities (Scott & Bruce, 1994). According to Amabile (1988):

These resources include a wide array of elements: people with knowledge of the feasibility of implementing particular innovations, people who have familiarity with relevant markets, people who have other types of relevant experience in the task domain, funds allocated to this work domain, material resources (such as existing means of production within the organization), systems of production, market research resources, data bases of relevant information, and the availability of personnel training in relevant ideas. (p. 154)

Idea development centers that go beyond the typical employee suggestion program by providing funding and technical support for the pursuit of new ideas are becoming increasingly common and have been used in companies such as Union Carbide, Bell Canada, Amoco Chemical, Eastman Kodak, American Greeting Cards, Du Pont, and the Body Shop (Gundry, Kickul, & Prather, 1994; Rosenfeld & Servo, 1990).

When employees see management and organizational policies and practices as protecting their welfare and interests in ways that allow them to experiment with new ideas, the organizational climate can be described as having a strong *socioemotional support* component. Socioemotional support from the larger work environment, conveyed by encouraging open and free debate and discussion, develops feelings of trust and freedom in employees (Ekvall & Tangeberg-Anderson, 1986; Mumford & Gustafson, 1988; West, 1990). Frischer (1993) points out that when managers' roles involve empowerment of subordinates, a climate is created that is supportive of creative pursuits. Subordinates then perceive that problems and conflicts are dealt with fairly and realistically, and will thus be more willing

to engage in the risk-taking behavior associated with creativity. Lack of internal competition or frequent and threatening evaluations can enhance perceptions that the organization is providing employees the support to freely engage in new ideas and projects.

Although we have focused attention on human resource practices in describing how organizational arrangements can facilitate a multidimensional climate for creativity, it is important to note that other organizational variables can influence climate perceptions. For instance, an organization's structure and physical layout can facilitate individual creativity and innovation (Kanter, 1988). Both may serve a specific role in the development of a creativity enhancing climate by signaling to employees how to be innovative (e.g., discuss ideas with others) and providing the support necessary for high levels of inventiveness. As another example, other organizational practices, such as accounting-related controls, can influence innovation (Tosi, 1992).

In summary, a number of different aspects of the work environment, particularly human resource practices, have been suggested as contributing to a climate for creativity. We have described how these types of practices may relate to particular dimensions of such a climate. Although considered separately here, the different dimensions of climate require coordination and consistency because climate perceptions are global and based on aggregate practices and procedures (Kopelman et al., 1990; Schneider et al., 1994). For example, if goals are set for creativity, yet the means to achieve those goals are not clearly outlined and the employees are not provided the task support necessary to achieve them, a mixed signal is sent. Thus, in order for a strong climate for creativity to exist, it is important that organizational structures, practices, policies, and procedures are designed to complement and be aligned with one another in supporting the creativity process.

NEW DIRECTIONS IN THE STUDY OF CULTURES AND CLIMATES FOR CREATIVITY AND INNOVATIONS

Culture and Climate Change

In this section we summarize our discussion of the literature and direct attention to potential new issues and directions in studying cultures and climates for creativity and innovation. This discussion is structured around three themes: (a) the ability of corporate cultures and climates to change so that they are more supportive of creativity, (b) the co-existence of multiple cultures and climates for creativity with other strategic imperatives, and (c) the relationship between culture and climate for creativity and innovation and quality management practices.

The heightened attention on creativity in work settings is in large part due to the hopes of improving organizational effectiveness. Developing a culture and climate supportive of creativity, however, requires a long and difficult process of change (Schneider et al., 1994). As we have suggested here, culture and climate for creativity are closely intertwined and mutually reinforcing. Attempts to change one without consideration of the other are likely to be futile. A double-barreled effort that is more accurately described as one of guided evolution than of change seems most appropriate. Research examining issues of what strategies are most effective for transitioning an organization's culture and climate to supporting creativity, the rate at which these changes can be accomplished, and the outcomes of these efforts, both in terms of individual creativity and organizational innovative performance and effectiveness, are clearly needed.

Because culture operates at a deeper level of meaning (i.e., the "why" that underlies patterns of behavior, practices, and routines that relate to creativity), involving long-established norms, values, and beliefs that are central to the organization, it is more difficult and slow to change than climate. In fact, it is more difficult to change an existing culture than it is to create a new one (Thompson & Luthans, 1990). This is one reason why a number of organizations have opted to open new plants with the types of characteristics including organization design, technology, managerial practices, and strategies that reflect a culture supportive of creativity rather than attempting to make dramatic transitions by changing the culture of existing plants. General Motors' Saturn Corporation is one well known example of this type of effort, but other organizations that have taken this approach include Mead, TRW, Sherwin Williams, Cummins Engine, and General Foods (Lawler, 1992).

Despite the fact that changing a culture from one based more on complacency and stability to one oriented toward change, risk taking, and inventiveness is difficult and slow, there are ways to make it happen. Visionary leadership, or the role of top management in defining a long-term vision of the organization based on concepts of creativity (Anderson et al., 1994), is one element important in changing an organization's culture. By specifying a new direction for the organization through a reorientation and convergence (Tushman & Romanelli, 1985) that is structured around assumptions and values of creativity and innovation, visionary leadership can provide a new role model for employees to follow. Schein (1990) outlines strategies that top management can use to direct the change toward a

new culture which include: (a) unfreezing existing assumptions and beliefs by pointing out threats to the organization if the status quo is maintained and emphasizing that change is possible and necessary, (b) rewarding new practices and behaviors that are consistent with the values and beliefs that are being encouraged (i.e., creativity and change), (c) filling key positions with new incumbents who share the new assumptions, and (d) establishing new rituals and symbols around the new set of assumptions.

Because organizational environments are a function of the persons behaving in them, the attraction, selection, and retention of certain kinds of people (i.e., those whose beliefs favor creativity, have a high tolerance for ambiguity, and value innovation and change) may be managed in a way to develop a workforce that places a high value on creativity and innovation and, hence, leads to the development of a creativity enhancing culture. Schneider (1987) theorized that organizations attract, select, and retain employees who tend to share similar values, personality traits, and cognitive styles. The attraction, selection, and retention of similar types of people has important implications for the development and evolution of culture. Over time, through these processes, those in the organization tend to share a similar cognitive style that influences how they analyze their work environment. By sharing similar views on why events and situations happen, people are expressing their shared attributions for what goes on in the organization. The myths and stories that are created become the essence of an organization's culture. As Schneider (1987) has aptly noted, in this sense, the people really do make the place.

It is important to note that Schneider's (1987) fundamental prediction is that, because (a) people are attracted to organizations they see as consistent with their own values and beliefs, (b) organizations choose individuals whose personal attributes more closely resemble those already in the organization, and (c) individuals who do not share these attributes tend to leave, organizations become composed of individuals who share a number basic values, beliefs, and personality traits. As a result, there is a natural tendency for organizations to become more homogeneous, ingrown, and less able to change over time. One immediate implication is that less diversity in the organization over time will result in lower creativity and innovation.

But this process might be turned on its head through active management of attraction, selection, and attrition. Recruitment of new employees should focus on attracting those with personal characteristics (i.e., knowledge, abilities, personality, motivational orientation, values, beliefs) associated with creativity and success in working in a highly innovative work environment. Selection strategies can be adapted to hire those who have (a) strong "domain-relevant skills" and "creativity-relevant skills" (Amabile, 1988), (b) high levels of cognitive abilities related to creativity such as flexibility, originality, and associative fluency (Woodman et al., 1993), and (c) personality traits and values such as high energy, autonomy, internal locus of control, and tolerance for ambiguity that are associated with creativity (Woodman et al., 1993).

A final way to manage the evolution of a new culture oriented toward creativity and innovation is through socializing new members. Using a set of seven basic dimensions along which socialization practices can vary, Van Maanen and Schein (1979) have argued that socialization and orientation practices can be designed so that they encourage a basic orientation either toward conformity with organizational norms and assumptions (i.e., custodial orientation) or acceptance of the basic and central assumptions, but rejection of the more peripheral ones (i.e., creative individualism). Developing an orientation toward creative individualism within the organization allows individuals to be more innovative in performing their tasks. By structuring socialization techniques in a way that encourages creative individualism (e.g., using informal methods such as coaching by the immediate supervisor, rather than in formal orientation sessions), it is possible to foster values and behaviors of creativity and innovation (Jones, 1986).

Through the types of employees that are recruited and selected and the ways in which new employees are socialized into the organization, the recruitment, selection, and socialization practices also send strong messages to those already in the organization by changing the organization's climate (Schneider et al., 1994). Adjusting an organizational climate so that it is more supportive of creativity and innovation is certainly difficult, but because climate is what employees perceive as emphasized in organizational structures, practices, procedures, and policies, the change process is more directly modifiable than it is for culture. It is important to again raise the point that climate is based on aggregate perceptions. Thus, change toward a climate for creativity and innovation requires coordinated adjustments across various human resource and organizational practices. Considering the types of practices that most directly relate to specific dimensions of climate, it is possible to produce a prescriptive set of activities (e.g., training, reward, leadership, resource support) that in combination produce a

strong aggregate climate for creativity that addresses the critical elements of climate: goals, means, rewards, task support, and socioemotional encouragement.

Multiple Cultures and Climates

The phrase "multiple cultures and climates" has two connotations. The first is that because events, actions, and practices differ across various units, divisions, and levels in a single organization and are often interpreted in a unique fashion by the people in them, multiple subcultures and climates may exist (Schneider, 1975; Thompson & Luthans, 1990). For example, Nystrom's (1990) analysis of four divisions of a Swedish chemical company revealed that each displayed a very different level of support for innovation in its culture and climate. One potential advantage of having multiple subcultures and climates operating within a single organizational entity is that it allows for building cultures and climates that support creativity and innovation in the areas of the organization where they are most necessary. This may result in an interaction between culture and climate creativity where an organization may use a deliberate strategy that isolates creativity in specific areas of the organization. For example, in R&D units high levels of creativity and innovation are more critical than in production, where continual reinvention of manufacturing methods and processes may actually serve to inhibit unit effectiveness.

The second way to consider multiple cultures and climates is in terms of the different orientations that are important to the organization. Culture and climate can represent a variety of different foci within the same organization. For instance, Schneider et al. (1994) argue that organizations such as 3M are so successful because they have been able to develop cultures and climates that emphasize the achievement of multiple priorities of innovation, service, and cooperation. Interesting issues that need further consideration along these lines are: (a) Within an overall organizational culture and climate for creativity and innovation, is it possible to successfully manage the "strength" and "orientation" of various subcultures and climates in a way that maximizes the needs of a particular unit or division? (b) If so, how can the configuration of various cultures and climate be accomplished?

Quality Management and Creativity

Creativity and innovation are often associated with the use of principles of quality management and continuous improvement (e.g., Adair & Murray, 1994; Brocka & Brocka, 1992). As the former CEO and chairman of 3M, A.F. Jacobson, stated, "Quality and innovation. These are the tools for staying ahead in our increasingly competitive society" (as quoted in Hiam, 1992, p. 159). Creativity is considered as integral to the concept of continuous improvement which is focused on striving for constant progress in product, process, and service quality. Creativity as considered in the context of total quality is best described as incremental innovation (Anderson et al., 1994). It involves a gradual, yet continual, process where changes are planned, tested, observed, and then implemented in an effort to achieve improved quality. This approach to creativity is closely associated with the concept of organizational learning (e.g., Argyris & Schon, 1978; Brown & Duguid, 1991) by encouraging the use of ongoing training and education as knowledge-seeking activities that provide the means for quality improvement. These types of organizational learning activities, according to Deming and other quality proponents, produce two types of knowledge: process task knowledge, the understanding of the processes that lead to the products and services that are produced, and profound task knowledge, the foundational knowledge of areas such as statistics, systems theory, and psychology necessary for investigating how to improve quality (Anderson et al., 1994). Variation is reduced as both profound and process knowledge are acquired and organizational learning advances through incremental innovation leading to improvements in processes, products, or services.

Thus, although creativity and innovation are central mechanisms in the quality and continuous improvement movement, they are managed in such a way as to minimize variation. The concept of incremental innovation is clearly different from the notion of radical change or a shift in paradigms. In fact, incremental innovation may actually serve to retard the development of decidedly new ideas, solutions, or products by focusing on minimizing variation in processes, products, and services. This may be one reason why Nystrom (1990) found that the most innovative division in his study also had a low orientation toward quality.

The distinction between incremental innovation and novel innovation in the total quality movement was noted by Sitkin, Sutcliff, and Schroeder (1994). They argue that there is an implicit duality in the goals of total quality management between a desire for control (and reduction in variation) and a desire for learning through creativity. Sitkin et al. (1994) argue that different types of total quality strategies that focus on one or the other of these goals are more appropriate in certain types of environments. For instance, in the auto industry, production processes are stable and characterized by

high volume runs, the use of well-known technologies, and routine tasks. In this type of environment creativity and innovation efforts should be directed at understanding existing customer needs and using statistical control and monitoring of the production process to make incremental changes that result in reduced variation. In contrast, in the computer industry, where rapidly evolving engineering advances mean that new models are introduced every 6 to 12 months, it is extremely difficult for organizations to put in place the standard quality systems that allow them to achieve the types of stable production process as seen in the auto industry. Moreover, under these conditions of rapid change, stringent quality control methods and standards that stress reducing errors can undermine the experimentation required to develop new products and introduce them into the market. In this type of environment, "quality" refers to the development of truly original products that can only be achieved through committing numerous errors and learning from failed experiments. In this light, different approaches to quality require different approaches to creativity. Research addressing these differing approaches to quality and how their corresponding cultures and climates support creativity and innovation is needed.

CONCLUSION In this paper we have attempted to integrate the concepts of organizational culture and climate into a framework that considers their development, relationship to organizational practices, and influences on individual creativity. Understanding the distinctions between culture and climate, as well as their interrelationships, should benefit both research and practice aimed at understanding how to structure work environments that support the creativity process. The deeply rooted values, beliefs, and patterns of behavior that compose an organization's culture require consistent changes in behaviors and establishing new meanings for these actions that are centered on values, beliefs, and assumptions of creativity and innovation. Climate may be adjusted through policies and practices, organizational structures, and the physical arrangements of work so that they are more consistent with and directly support the critical dimensions of a climate for creativity and innovation. The framework we have presented suggests that although culture is more slow and difficult to change than climate, because climate is ultimately a manifestation of culture, the long-term success of efforts to develop organizational conditions that support creativity and innovation requires the use of strategies that are targeted at both culture and climate. Our model and related ideas are necessarily speculative at this time due to the relative lack of existing empirical research. We hope that this article will stimulate additional thinking and research on the effects of organizational culture and climate on creativity in work settings.

REFERENCES ABBEY, A., & DICKSON, J.W. (1983). R&D work climate and innovation in semi-conductors. *Academy of Management Journal*, 26, 362-368.

- ADAIR, C.B., & MURRAY, B.A. (1994). *Break-through process redesign: New ways to customer value*. New York: AMACOM.
- AMABILE, T.M. (1988). A model of creativity and innovation in organizations. *Research in Organizational Behavior*, 10, 123-167.
- AMABILE, T.M. & GRYSKIEWICZ, S.S. (1987). *Creativity in the R&D laboratory*. Greensboro, NC: Center for Creative Leadership.
- ANDERSON, J.C., RUNGTUSANATHAM, M., & SCHROEDER, R.G. (1994). A theory of quality management underlying the Deming management method. *Academy of Management Review*, 19, 472-509.
- ANDERSON, N., & KING, N. (1993). Innovation in organizations. In C.L. Cooper & I.T. Robertson (Eds.), *International review of organizational psychology*, (Vol 8). Chichester: Wiley.
- ARGYRIS, C., & SCHON, D. (1978). *Organizational learning: A theory of action perspective*. Reading, MA: Addison-Wesley.
- BANDURA, A. (1986). *Social foundations of thought and action*. Englewood Cliffs, NJ: Prentice-Hall.
- BROCKA, B., & BROCKA, M.S. (1992). *Quality management: Implementing the best ideas of the masters*. Homewood, IL: Irwin.
- BROWN, J., & DUGUID, P. (1991). Organizational learning and communities of practice: Toward a unified view of working, learning, and innovation. *Organizational Science*, 2, 40-57.
- CAMPBELL, J.P., DUNNETTE, M.D., LAWLER, E.E. III, & WEICK, K.E. (1970). *Managerial behavior, performance, and effectiveness*. New York: McGraw-Hill.
- CHATMAN, J.A. (1991). Matching people and organizations: Selection and socialization in public accounting firms. *Administrative Science Quarterly*, 36, 459-484.
- CIOTTA, P.J. (1987). The anatomy of a creative corporate culture. *Journal of Creative Behavior*, 21(2), 145-152.
- COOKE, R.A., & ROUSSEAU, D.M. (1988). Behavioral norms and expectations: A quantitative approach to the assessment of organizational culture. *Group and Organization Studies*, 13, 245-273.
- COUGER, J.D., HIGGINS, L.F., & MCINTYRE, S.C. (1993). Unstructured creativity in information systems organizations. *MIS Quarterly*, 375-397.
- EKVALL, G., & TANGEBERG-ANDERSSON, Y. (1986). Working climate and creativity: A study of an innovative newspaper office. *Journal of Creative Behavior*, 20(3), 215-225.
- EREZ, M. (1992). Interpersonal communication systems in organizations, and their relationships to culture values, productivity and innovation: The case of Japanese corporations. *Applied Psychology: An International Review*, 41, 43-64.

- FARR, J.L., & FORD, C.M. (1990). Individual innovation. In M.A. West & J.L. Farr, (Eds.), *Innovation and creativity at work: Psychological and organizational strategies* (pp. 63-80). New York: Wiley.
- FRISCHER, J. (1993). Empowering management in new product development units. *Journal of Product Innovation Management*, 10, 393-401.
- GORDON, G.G. (1991). Industry determinants of organizational culture. *Academy of Management Review*, 16, 396-415.
- GUNDRY, L.K., KICKUL, J.R., & PRATHER, C.W. (1994). Building the creative organization. *Organizational Dynamics*, 22(4), 22-37.
- GUZZO, R.A. (1988). Productivity research: Reviewing psychological and economic perspectives. In J.P. Campbell, R.J. Campbell, & Associates, *Productivity in organizations: New perspectives from industrial and organizational psychology* (pp. 63-81). San Francisco: Jossey-Bass.
- HAGE, J., & DEWAR, R. (1973). Elite values versus organizational structure in predicting innovation. *Administrative Science Quarterly*, 18, 279-290.
- HARRIS, D.H. (1994). *Organizational linkages: Understanding the productivity paradox*. Washington, D.C.: National Academy Press.
- HIAM, A. (1992). *Closing the quality gap: Lessons from America's leading companies*. Englewood Cliffs, NJ: Prentice Hall.
- JONES, G.R. (1986). Socialization tactics, self-efficacy, and newcomers' adjustment to organizations. *Academy of Management Journal*, 29, 262-279.
- KABANOFF, B., & ROSSITER, J.R. (1994). Recent developments in applied creativity. In C.L. Cooper & I.T. Robertson, (Eds.), *International review of industrial and organizational psychology*, Vol. 9. Chichester: Wiley.
- KANTER, R.M. (1988). When a thousand flowers bloom: Structural, collective, and social conditions for innovation in organizations. *Research in Organizational Behavior*, 10, 169-211.
- KERR, J. & SLOCUM, J.W. (1987). Managing organizational culture through reward systems. *Academy of Management Executive*, 1, 99-108.
- KING, N., & ANDERSON, N. (1990). Innovation in working groups. In M.A. West & J.L. Farr, (Eds.), *Innovation and creativity at work: Psychological and organizational strategies*. New York: Wiley.
- KOPELMAN, R.E., BRIEF, A.P., & GUZZO, R.A. (1990). The role of climate and culture in productivity. In B. Schneider (Ed.), *Organizational climate and culture* (pp. 282-318). San Francisco: Jossey-Bass.
- KOZLOWSKI, S., & DOHERTY, M. (1989). Integration of climate and leadership: Examination of a neglected topic. *Journal of Applied Psychology*, 74: 546-553.
- LAWLER, E.E. III (1992). *The ultimate advantage: Creating the high-involvement organization*. San Francisco: Jossey-Bass.
- LOEB, M. (1995). Ten commandments for managing creative people. *Fortune*, 131(1), 135-136.
- LOCKE, E.A., SHAW, K.N., SAARI, L.M., & LATHAM, G.P. (1981). Goal setting and task performance: 1969-1980. *Psychological Bulletin*, 90, 125-152.
- LOUIS, M.R. (1980). Surprise and sensemaking: What newcomers experience in entering unfamiliar organizational settings. *Administrative Science Quarterly*, 25, 226-251.
- MARTIN, J., FELDMAN, M., HATCH, M., & SITKIN, S. (1983). The uniqueness paradox in organizational stories. *Administrative Science Quarterly*, 28, 438-453.
- MEYER, A.D. (1982). Adopting to organizational jolts. *Administrative Science Quarterly*, 27, 515-537.
- MUMFORD, M.D., & GUFTASON, S.B. (1988). Creativity syndrome: Integration, application and innovation. *Psychological Bulletin*, 103, 27-43.
- NONAKA, I. (1991). The knowledge-creating company. *Harvard Business Review*, 69, 96-104.
- NYSTROM, H. (1990). Organizational innovation. In M.A. West & J.L. Farr, (Eds.), *Innovation and creativity at work: Psychological and organizational strategies* (pp. 143-162). New York: Wiley.
- PAYNE, R. & PUGH, D.S. (1976). Organizational structure and climate. In M.D. Dunnette (Ed.), *Handbook of industrial and organizational psychology* (pp.1125-1173). Chicago: Rand McNally.
- PELTZ, D., & ANDREWS, P.S. (1966). *Scientists in organizations*. New York: Wiley.
- PIERCE, J.L., & DELBEQUE, A. (1977). Organizational structure, individual attitudes, and innovation. *Academy of Management Review*, 2, 27-37.
- PORTER, L.W., & LAWLER, E.E. (1968). *Managerial attitudes and performance*. Homewood, IL: Irwin-Dorsey Press.
- REDMOND, M.R., MUMFORD, M.D., & TEACH, R. (1993). Putting creativity to work: Effects of leader behavior on subordinate creativity. *Organizational Behavior and Human Decision Processes*, 55, 120-151.
- REICHERS, A.E., & SCHNEIDER, B. (1990). Climate and culture: An evolution of constructs. In B. Schneider (Ed.), *Organizational climate and culture*. San Francisco: Jossey-Bass.
- ROSENFELD, R., & SERVO, J.C. (1990). Facilitating innovation in large organizations. In M.A. West & J.L. Farr, (Eds.), *Innovation and creativity at work: Psychological and organizational strategies* (p.251-264). New York: Wiley.
- ROUSSEAU, D.M. (1990). Assessing organizational culture: The case for multiple methods. In B. Schneider (Ed.), *Organizational climate and culture*. San Francisco: Jossey-Bass.
- SCHEIN, E.H. (1985). *Organizational culture and leadership*. San Francisco: Jossey-Bass.
- SCHEIN, E.H. (1990). Organizational culture. *American Psychologist*, 45, 109-119.
- SCHNEIDER, B. (1975). Organizational climates: An essay. *Personnel Psychology*, 28, 447-479.
- SCHNEIDER, B. (1987). The people make the place. *Personnel Psychology*, 40, 437-453.

- SCHNEIDER, B. (1990). The climate for service: An application of the climate construct. In B. Schneider (Ed.), *Organizational climate and culture* (pp. 383-412). San Francisco: Jossey-Bass.
- SCHNEIDER, B., & BOWEN, D.E. (1985). Employee and customer perceptions of customer service in banks: Replication and extension. *Journal of Applied Psychology*, 70, 423-433.
- SCHNEIDER, B., GUNMARSON, S.K., & NILES-JOLLY, K. (1994). Creating the climate and culture of success. *Organizational Dynamics*, 23(1), 17-29.
- SCHNEIDER, B., & RENTSCH, J. (1988). Managing climates and cultures: A futures perspective. In J. Hage (Ed.), *Futures of organizations* (pp. 181-200). Lexington, MA: Lexington Books.
- SCHULER, R.S., & JACKSON, S.E. (1987). Linking competitive strategies with human resource management practices. *Academy of Management Executive*, 1, 207-219.
- SCOTT, S.G., & BRUCE, R.A. (1994). Determinants of innovative behavior: A path model of individual innovation in the workplace. *Academy of Management Journal*, 37, 580-607.
- SITKIN, S.B., SUTCLIFFE, K.M., & SCHROEDER, R.G. (1994). Distinguishing control from learning in total quality management: A contingency perspective. *Academy of Management Review*, 19, 537-564.
- THOMPSON, K.R., & LUTHANS, F. (1990). Organizational culture: A behavioral perspective. In B. Schneider (Ed.), *Organizational climate and culture* (pp. 319-344). San Francisco: Jossey-Bass.
- TORNATZKY, L.G., FKEISHER, M., & CHALRABARTO, A.K. (1990). *The process of technological innovation*. Lexington, MA: Lexington Books.
- TOSI, H.L. (1992). *The environment/organization/person contingency model: A meso approach to the study of organizations*. Greenwich, CT: JAI Press.
- TRICE, H.M., & BEYER, J.M. (1984). Studying organizational cultures through rites and ceremonies. *Academy of Management Review*, 9, 653-669.
- TUSHMAN, M.L., & ROMANELLI, E. (1985). Organizational evolution: A metamorphosis model of convergence and reorientation. In L.L. Cummings & B.M. Staw (Eds.), *Research in organizational behavior* (Vol. 7), pp. 171-222. Greenwich, CT: JAI Press.
- VAN MAANEN, J., & SCHEIN, E.H. (1979). Toward a theory of organizational socialization. In B.M. Staw & L.L. Cummings (Eds.), *Research in organizational behavior* (Vol. 1), pp. 204-264. Greenwich, CT: JAI Press.
- VERESPEJ, M.A. (1994). Managing for creativity. *Industry Week*, 244(8), 24-26.
- VROOM, V.H. (1964). *Work and motivation*. New York: Wiley.
- WEST, M.A. (1990). The social psychology of innovation in groups. In M.A. West & J.L. Farr, (Eds.), *Innovation and creativity at work: Psychological and organizational strategies* (pp. 309-334). New York: Wiley.
- WOODMAN, R., SAWYER, J., & GRIFFIN, R. (1993). Toward a theory of organizational creativity. *Academy of Management Review*, 18, 293-321.
- ZOHAR, D. (1980). Safety climate in industrial organizations: Theoretical and applied implications. *Journal of Applied Psychology*, 65, 96-102.

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